

LI-LI YE

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EDUCATION AND WORK EXPERIENCE

Arizona State University, Tempe, Arizona, USA 2026 – Present

- Postdoctoral Researcher, Working in [Dr. Ying-Cheng Lai](#)'s Lab
- Research Focus: Quantum Reservoir Computing and AI for Laser Systems.

Arizona State University, Tempe, Arizona, USA 2021 – 2026

- Ph.D. in Electrical Engineering, *Advisor*: [Dr. Ying-Cheng Lai](#)
- Research Focus: Quantum Physics and Machine Learning.
- [Dean's Dissertation Award](#).

Lan-Zhou University, Cheng-Guan, Gan-Su, China 2020 – 2021

- M.S. in Theoretical Physics, *Advisor*: [Dr. Liang Huang](#)

Lan-Zhou University, Cheng-Guan, Gan-Su, China 2016 – 2020

- B.S. in Theoretical Physics
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PEER-REVIEWED PUBLICATIONS

C15. N. Vigne , A. Pando, **L.-L. Ye**, M. Bütün, Y.-C. Lai, N. Davidson F.-Y. Lin, and H. Cao. “**Tailoring laser coupling for efficient synchronization.**”

Conference: Semiconductor Lasers and Laser Dynamics XII (2026).

J14. L.-L. Ye, N. Vigne, F.-Y. Lin, H. Cao, and Y.-C. Lai. "**Disorder-mediated synchronization resonance in coupled semiconductor lasers.**" Physical Review Research 8(1), 013104, (2026).

J13. J.-Y. Huang, Z.-M. Zhai, L.-L. Ye, V. Kovanis, Y.-C. Lai. "**Optimizing disorder with machine learning to harness phase synchronization.**" Chaos: An Interdisciplinary Journal of Nonlinear Science, 36(4), (2026).

J12. L.-L. Ye, C. Arenz, J. M. Lukens, and Y.-C. Lai. "**Entanglement engineering of optomechanical systems by reinforcement learning.**" APL Machine Learning, 3(1), (2025).

J11. L.-L. Ye and Y.-C. Lai. "**Controlling nonergodicity in quantum many-body systems by reinforcement learning.**" Physical Review Research, 7(1), 013256, (2025).

J10. J.-Y. Huang, L.-L. Ye, and Y.-C. Lai, "**Floquet quantum many-body scars in the tilted Fermi-Hubbard chain.**" Physical Review B 112(1), 014303, (2025).

J9. L.-L. Ye, C.-D. Han, and Y.-C. Lai. "**Optical properties of two-dimensional Dirac-Weyl materials with a flatband.**" Applied Physics Letters, 124(6), (2024).

J8. L.-L. Ye, C.-Z. Wang, and Y.-C. Lai. "**Experimental scheme for determining the Berry phase in two-dimensional quantum materials with a flat band.**" Physical Review B, 110(7), 075108, (2024).

J7. S. Panahi, L.-L. Ye, and Y.-C. Lai. "**Higher-order exceptional points in noise-assisted sensing structure.**" Physical Review Applied 22 (5), 054063, (2024).

J6. C.-D. Han, L.-L. Ye, Z. Lin, V. Kovanis, and Y.-C. Lai. "**Deep-learning design of graphene metasurfaces for quantum control and Dirac electron holography.**" APL Machine Learning 2, 036105, (2024).

J5. L.-L. Ye and Y.-C. Lai. "**Irregular Bloch-Zener oscillations in two-dimensional flat-band Dirac materials.**" Physical Review B, 107(16), 165422, (2023).

J4. L.-L. Ye, C.-D. Han, and Y.-C. Lai. "**Spin-dependent edge states in two-dimensional Dirac materials with a flat band.**" Physical Review B, 108(23), 235404, (2023).

C3. M.-H. Guo, Y. Weng, L.-L. Ye, and Y.-C. Lai. "**Continuous variational quantum algorithms for time series.**" 2023 International Joint Conference on Neural Networks (IJCNN), 01-08, (2023).

J2. L.-L. Ye, C.-D. Han, and Y.-C. Lai. "**Geometry-induced wave-function collapse.**" Physical Review A 106 (2), 022207, (2022).

J1. Z.-Y. Li, L.-L. Ye, R.-H. Ni, C.-Z. Wang, L. Huang, Y.-C. Lai, C. Grebogi. "**Relativistic quantum scarring, spin-induced phase, and quantization in a symmetric Dirac billiard system.**" Journal of Physics A: Mathematical and Theoretical. 55 (37), 374003, (2022).

AVAILABLE PREPRINTS

M5. L.-L. Ye, and Y.-C. Lai. "**Reservoir Computing: From Classical Edge of Chaos to Quantum Dynamical Phases.**" (accepted by 2026 Conference on Physics and AI (PAI26))

M4. Z.-M. Zhai, L.-L. Ye, C. Grebogi, and Y.-C. Lai, "**Machine learning with small data for nonlinear dynamics.**" Preprint

M3. L.-L. Ye, and Y.-C. Lai. "**Physics-Informed Network Optimization for Taming 1,000 Disordered Lasers.**" Preprint

M2. L.-L. Ye, N. Vigne, F.-Y. Lin, H. Cao, and Y.-C. Lai. "**Optimal sparse networks for synchronization of semiconductor lasers.**" (2026)

<https://arxiv.org/abs/2511.03205>.

M1. M. Moradi, L.-L. Ye, and Y.-C. Lai. "**Reinforcement Learning in Physics: A Survey.**" Preprint

HONORS AND AWARDS

- [Dean's Dissertation Award](#) of the Ira A. Fulton Schools of Engineering,

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| (The award is presented to up to 5% of the number of prior year's Ph.D. graduates) | 2026 |
| • APS GDS Travel Award (\$500) | 2026 |
| • ASU GSG Individual Travel Grant Program (\$950) | 2026 |
| • Graduate College Completion Fellowship (\$22120) | 2025 - 2026 |
| • Electrical Engineering Department Scholarship (\$1045) | 2025 |
| • APS DAMOP Participation Award (\$600) | 2025 |
| • ASU GSG Individual Travel Grant Program (\$950) | 2025 |
| • Engineering Graduate Fellowship (\$1000) | 2025 |
| • APS 4CS Annual Meeting Travel Award (\$950) | 2023 |
| • National Scholarship (rank 1/300) (\$3428) | 2016 - 2019 |

TALKS

- 2026 Conference on Physics and AI (PAI26) | Reservoir Computing: From Classical Edge of Chaos to Quantum Dynamical Phases (Poster)
 - 2026 APS March Meeting | Quantum Reservoir Computing for Noisy, High-Dimensional Magnetic Navigation (Talk)
 - 2026 APS March Meeting | Machine Learning Applications in Complex and Quantum Systems (Poster)
 - 2025 APS Conference for Graduate Women & Gender Minorities in Physics | Machine Learning Applications in Complex and Quantum Systems (Poster)
 - 2025 APS DAMOP | Entanglement engineering of optomechanical systems by reinforcement learning (Talk)
 - 2025 SIAM Conference on Applications of Dynamical Systems | Disorder-induced synchronization in laser models (Talk)
 - 2025 APS March Meeting | Entanglement engineering of optomechanical systems by reinforcement learning (Talk)
 - 2023 APS Four Corners Meeting | Irregular Bloch-Zener Oscillations in 2D Flat-Band Dirac Materials (Talk)
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PROFESSIONAL SERVICE & SKILLS

- **Journal Referee:** Nature Communications, PRX Quantum, Physical Review A/B/E/Research/Applied, Annals of Physics (2023–Present).
- **Session Chair:** APS DAMOP (2025), APS March Meeting (2025), APS 4CS Meeting (2023).
- **Technical Skills:** Qiskit, QuTiP, Python (TensorFlow/Keras), Matlab, Mathematica, C++, LaTeX.